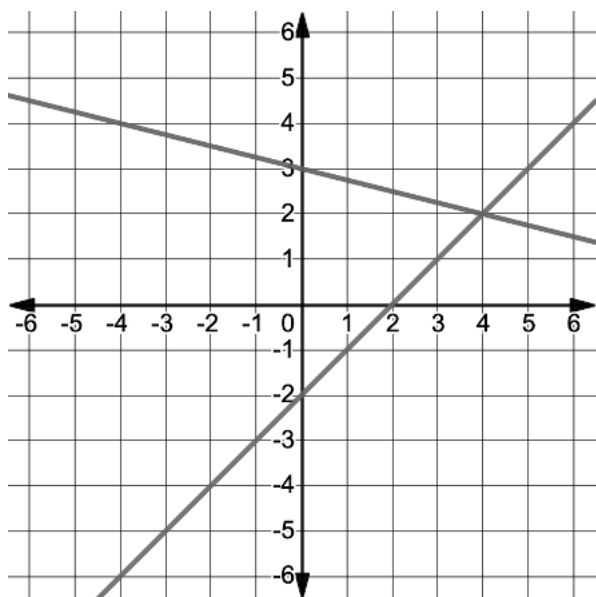


Grade 8
Unit 8
Lesson 7 Practice

Name Answer Key
Date _____
Period _____

The equations $y = -\frac{1}{4}x + 3$ and $y = x - 2$ are shown on the coordinate grid below.



1. Based on the graph, what is the solution to the system of equations $\begin{cases} y = -\frac{1}{4}x + 3 \\ y = x - 2 \end{cases}$?
(4,2)

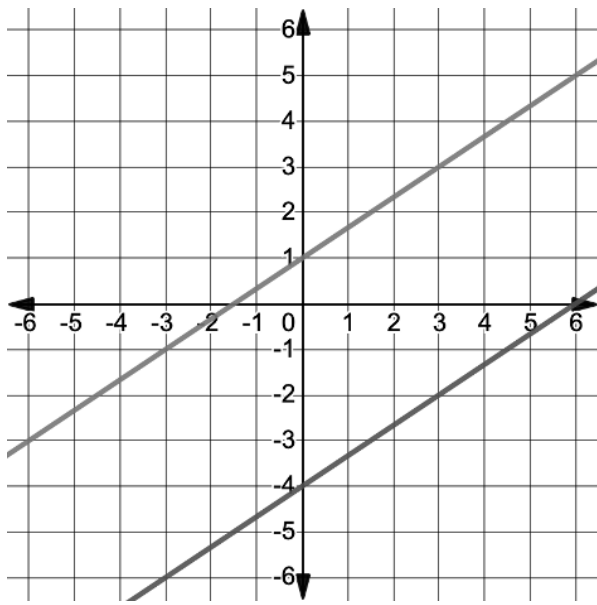
2. Solve the equation $-\frac{1}{4}x + 3 = x - 2$
 $-\frac{1}{4}x + 3 = x - 2$
 $-\frac{1}{4}x + 5 = x$
 $5 = \frac{5}{4}x$
 $x = 4$

3. How are your answers to questions 1 and 2 related?
The x -value of the point of intersection is the solution to the equation.

4. Use substitution to verify that your solution from question 1 satisfies both equations $y = -\frac{1}{4}x + 3$ and $y = x - 2$.

$y = -\frac{1}{4}x + 3$	$y = x - 2$
$2 = -\frac{1}{4}(4) + 3$ $2 = -1 + 3$ $2 = 2 \checkmark$	$2 = 4 - 2$ $2 = 2 \checkmark$

The system of equations $\begin{cases} y = \frac{2}{3}x + 1 \\ y = \frac{2}{3}x - 4 \end{cases}$ is shown on the coordinate grid below.



5. Solve the equation $\frac{2}{3}x + 1 = \frac{2}{3}x - 4$.

$$\frac{2}{3}x = \frac{2}{3}x - 5$$

$$0 \neq 5$$

No solution

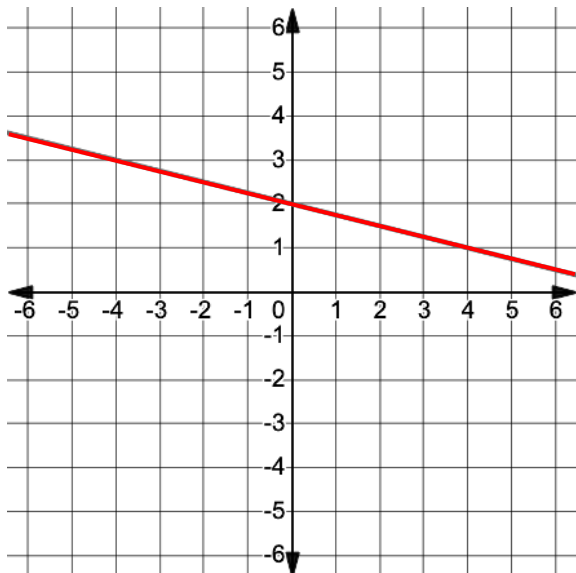
6. What is the solution to the system of equations $\begin{cases} y = \frac{2}{3}x + 1 \\ y = \frac{2}{3}x - 4 \end{cases}$?

No solution

7. Explain how the graph is related to the solution to the equation $\frac{2}{3}x + 1 = \frac{2}{3}x - 4$.

When there are no points of intersection on a graph, there will also be no solution to the system of equations.

The equation $y = -\frac{1}{4}x + 2$ is shown on the coordinate grid below.



8. Graph the equation $y = -0.25x + 2$.

9. What do you notice about the graphs of the equations $y = -\frac{1}{4}x + 2$ and $y = -0.25x + 2$?

They overlap each other perfectly.

10. Describe the solution(s) to the system of equations $\begin{cases} y = -\frac{1}{4}x + 2 \\ y = -0.25x + 2 \end{cases}$.

All the points on both lines are solutions to the system of equations. This means there are an infinite number of solutions.